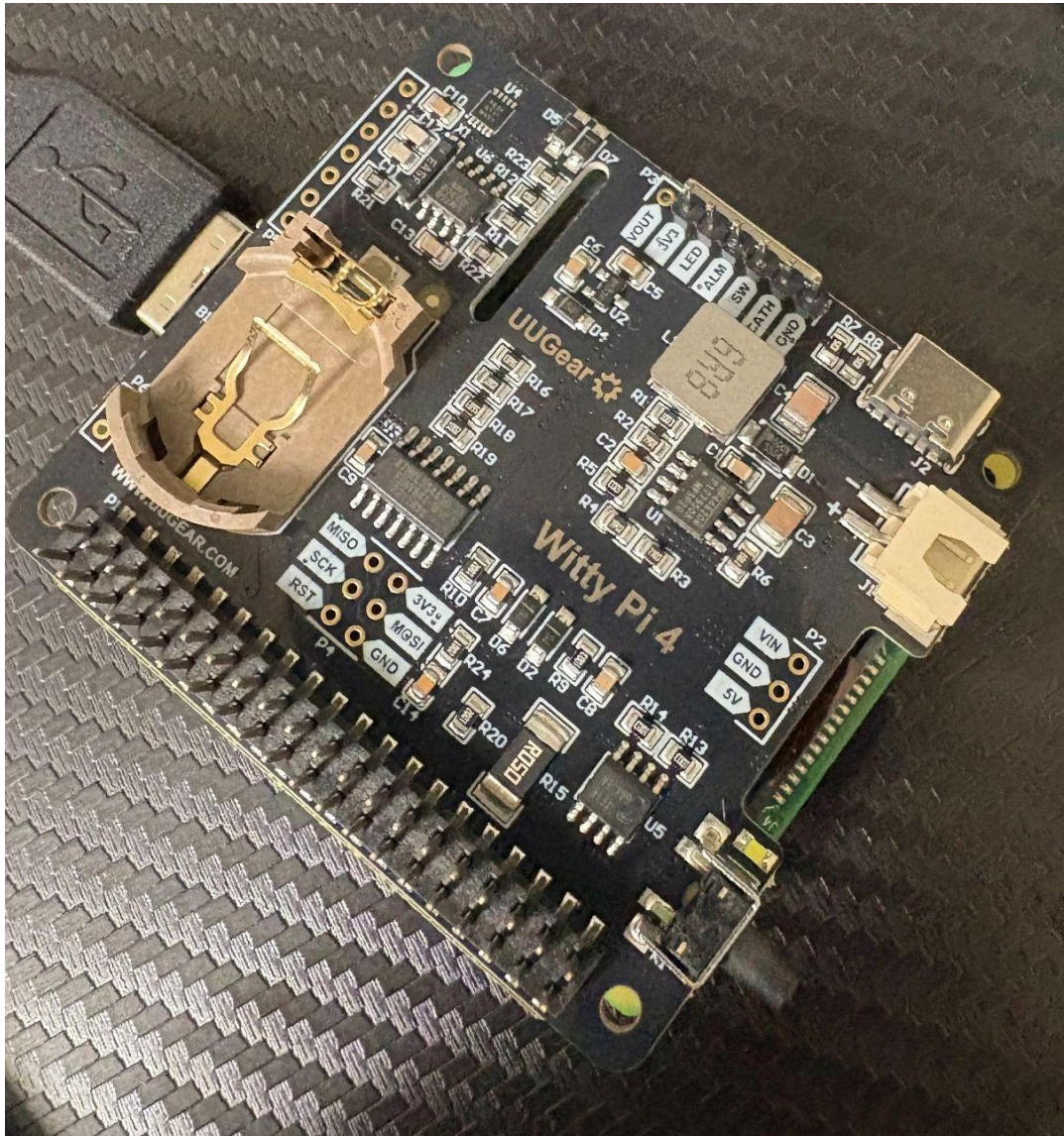


Software Build Guide



Required Materials:

Within the EDNA Sampler, but necessary if starting from scratch

Raspberry pi 3 A+

Witty Pi 4 (full size hat)

microSD card and adapter for computer

laptop/computer

Power cable (usb-c), and an outlet block

Extra monitor, keyboard, mouse, and hdmi cord

Step 1: Flashing the MicroSD and turning on the Raspberry Pi

1. Put the microSD card (using the adapter) into your laptop/computer
2. Google "Raspberry Pi Imager" on your laptop, and follow the instructions to download the imager onto your microSD card. You should be flashing the following option for Bullseye from raspberry pi's archives:
https://downloads.raspberrypi.com/raspios_armhf/images/raspios_armhf-2023-05-03/
Just download the first link. Should end in .img.xz.
3. Using the Raspberry Pi Imager (requires download), select rasp pi 3, select "other image from computer", and select the file you just downloaded (aka the 32 bit Bullseye img file). Select your SD card for your storage device, and select write.
4. After the img writes to your SD card, safely eject the SD card, and then remove the microSD, and put the microSD card into your Raspberry Pi 3 A+. It should be on the "underside" of the board, off to the side.
5. While the pi is still off, put on the Witty pi hat. Use the usb c on the Witty board for power supply, as opposed to the rasp pi itself (would use a micro USB cable). [If using the battery to power the Witty board, this may be fine too]
6. Plug your Rasp Pi into the hdmi cord/monitor, and your keyboard/mouse into the usb-b slot. You will only be able to use either the keyboard or mouse at a time without an adapter, but almost everything can be done with the keyboard alone. Now Turn on the Raspberry Pi by hitting the button!

Step 2: Setting up the Raspberry Pi (Witty software, EDNA_Sampler software, venv setup)

Before we begin this step I have some notes:

- While you are using Witty Pi, try to never unplug the power cord when the raspberry pi is processing (green LED is lighting up). Just run "sudo poweroff" in your terminal, or hit the button on the side of the witty Pi 4 to trigger a safe shutdown or power-on.
- If either shutdown method does not fully work (green turns off, but red light stays on), it is likely due to hdmi cord "backpowering", so unplugging your hdmi cord will likely solve the problem, and the red light should then turn off. However, as long as the green LED is off it is safe to unplug the usb-c cord from the Witty Pi board and it should not damage your SD card.
- Please note that the system is currently configured to have 6 specific pins (listed in [motors.py](#)) connected to the motors, and the LED GPIO pin connected to pin 11 (controlled in [Led.py](#)). Don't expect your system to run properly if you're not in the correct pin locations.

Now for the setup:

1. Set your Country, Language, Timezone to whatever yours is (Hawaii, USA for example). Then hit next.

2. Set an easy username/password (for example, user:ore653, pass:ORE653). Wifi can be set up, hit SKIP regarding the software update, and restart.
 - a. Regarding wifi, ideally use one that doesn't have a firewall. Personal Hotspot from your phone might be your best option if you are working from a University office. If you use your personal hotspot, enable maximum compatibility(on iphone).
 - b. Once at the main screen/desktop, on the keyboard, hit ctrl-alt-t to open up the terminal. Run: `sudo apt update && sudo apt upgrade -y`. This will update your system. NOTE: This will likely take a while.
3. Now for the Witty software download.
Run: `wget https://www.uugear.com/repo/WittyPi4/install.sh && sudo sh install.sh .`
4. Run: `sudo raspi-config`. Go to interface options, and enable SSH. Then enable I2C. Then hit Finish.
5. Run: `sudo reboot` (this will turn off the pi). Once it comes back up, hit ctrl-alt-t again to pull up the terminal.
6. Wait for wifi to reconnect. Then Run: `hostname`. Then run: `hostname -I`. These will give you your IP address (The first set of numbers). Keep these handy!
7. Run: `cd ~/Desktop`. Then run: `git clone https://github.com/Wredford/EDNA\_SAMPLER.git`
8. Run: `cd EDNA_SAMPLER`.
9. Next, you'll need to create a venv, which we'll call `edna_env`. Run the following:
`python3 -m venv edna_env`. Then run: `source edna_env/bin/activate`
10. Before moving forward, run: `date`. If your date and time are not correct, run: `sudo timedatectl set-ntp true`. Wait 10 seconds, and your date should become correct.
11. Next, you'll run the following from inside your venv, which you should now see in your command line: `sudo apt install ca-certificates -y`. Then run: `sudo update-ca-certificates`
12. Run: `python -m ensurepip --upgrade`. Then run: `python -m pip install --upgrade pip setuptools wheel`
13. Now run:


```
pip install numpy
pip install flask
pip install adafruit-blinka
sudo apt install python3-rpi.gpio
pip install RPi.GPIO
```
14. Now to give yourself permission to write the schedule! Run: `sudo chown ore653 /home/ore653/wittypi/schedule.wpi`. If you didn't name your pi ore653, change accordingly.

Now we can try to run the code using an existing wifi network! If you run: `ls`. You should see a few .py scripts, including `app.py`. Now run: `python app.py`. On another device connected to the same wifi as the rasppi (can be a phone, laptop, computer, etc), open up one of the http links that the terminal shows. From here, you can try to save some user

inputs. If it goes to another page, and then allows you to hit “back”, then it’s probably working fine. You can test other buttons as well, but note that hitting “prepare for next deployment” will turn off the raspberry pi. Not ideal if you're still trying to set it up.

Step 3: Having the Raspberry Pi act as its own router, and run the app on startup

Note: These next steps will make the pi into its own router, making it so that the pi can no longer access wifi. It can be changed back by commenting out the text you will add to `dhcpcd.conf` in a moment.

1. Run: `sudo apt install hostapd dnsmasq -y`. Then run: `sudo systemctl unmask hostapd`. Then run: `sudo systemctl enable hostapd`.
2. Run: `sudo nano /etc/dhcpcd.conf`. Then paste/type this at the bottom (You can comment this out if you ever need to go back to using the wifi):

```
interface wlan0
static ip_address=192.168.4.1/24
nohook wpa_supplicant
```

To save, hit `ctrl + X`, then `y`. There should be controls listed on the bottom of the nano editor

3. Now we will configure DHCP. Run: `sudo mv /etc/dnsmasq.conf /etc/dnsmasq.conf.orig`. Then run: `sudo nano /etc/dnsmasq.conf`. Paste or type in the following:

```
interface=wlan0
dhcp-range=192.168.4.2,192.168.4.20,255.255.255.0,24h
```

4. Now we will configure the “wifi network” coming from the Raspberry Pi. Run: `sudo nano /etc/hostapd/hostapd.conf`. Paste in the following:

```
interface=wlan0
driver=nl80211
ssid=EDNA_SAMPLER
hw_mode=g
channel=6
wmm_enabled=0
```

```
wpa=2
wpa_passphrase=edna1234
```

5. Run: `sudo nano /etc/default/hostapd`. Set `DAEMON_CONF="/etc/hostapd/hostapd.conf"`. Save with `Ctrl+X`, then `y`.
6. In your terminal, run: `sudo systemctl restart hostapd`. Then run: `sudo reboot`.
7. Now we'll make the raspberry Pi run the app on startup! Run: `sudo nano /etc/systemd/system/edna.service`. PASTE/TYPER IN THE FOLLOWING (make sure the working directory and `execStart` paths are correct for where you have your `EDNA_SAMPLER` folder located. We use username `ore653`):

[Unit]

Description=EDNA Sampler Service

After=network.target

[Service]

User=ore653

WorkingDirectory=/home/ore653/Desktop/EDNA_SAMPLER

ExecStart=/home/ore653/Desktop/EDNA_SAMPLER/edna_env/bin/python3

/home/ore653/Desktop/EDNA_SAMPLER/app.py

Restart=always

Environment=PYTHONUNBUFFERED=1

ExecStartPre=/bin/sleep 10

[Install]

WantedBy=multi-user.target

8. Hit CTRL + O to save, Enter, then CTRL + X to exit the editor.
9. Now enable the service! Run: `sudo systemctl daemon-reload`. Then run: `sudo systemctl enable edna.service`. Then run: `sudo systemctl start edna.service`.
10. Check if it's working! Run: `sudo systemctl status edna.service`. Then try to connect with your phone or another device. Should appear as a "EDNA_SAMPLER". Password will be edna1234, or whatever you have it set as. (EDNA2026 is password on current instrument). Now see if you can log into the app on your phone using <http://192.168.4.1:5000>.
11. A super useful tool for checking if your app is working is by running the following in the terminal on your rasp pi: `journalctl -u edna.service -f`. This will give you a log of all the events occurring, including the witty pi scheduler. You can use ctrl + C to exit back to terminal.